Adults' Experiences and Perceptions of Health Apps and Wearables: A Qualitative Analysis of Motivations and Sustained Use

17 November 2025

Submitted in partial fulfilment of the requirements for SPH 5409 Qualitative Methods in Public Health module

CONTENTS

Abstract 1
Introduction 1
Methods 2
Results 3
Discussion 6
Conclusion 8
References 9
AppendixA: Coding Structure 10
Appendix B: Codebook from Atlas.ti 11
Appendix C: COREQ Checklist 19

ABSTRACT

Background

Health trackers are ubiquitous; however, sustained use is an ongoing challenge. Thus, it is important to understand adults' lived experiences and perceptions of using health trackers, which includes systematically exploring motivations that lead to initial adoption and sustained use.

Aim

This study seeks to explore adults' experiences, perceptions, and motivations before, during, and after adopting a health tracker, with specific emphasis on how motivations evolve over time and across different types of users.

Methods

Qualitative thematic analysis of five in-depth interviews with adult health tracker users in Singapore. A hybrid deductive-inductive approach to utilising predetermined themes from behaviour change literature but also considered themes that emerged during the research study. Data analysis was undertaken with the support of Atlas.ti software.

Results

Five different domains of motivation emerged throughout the study: external incentives, gamified goals, social motivation, self-quantification, and barriers. In general, motivations evolved from external rewards to habitual behaviours. Though trackers were effective, they typically strengthened motivation for a pre-existing behavioural pattern rather than instigating an entirely new one. Additionally, effectiveness was highly contingent on user types and types of barriers involved in data collection. For instance, users with active barriers that required ongoing monitoring and effort (i.e., entering data manually every time) abandoned the tracker, while users with passive barriers (i.e., occasional inaccuracies) tolerated the tracker.

Conclusion

Health tracker engagement involves multiple, layered motivations that evolve temporally. Trackers function as optimisation tools for motivated users rather than behaviour change interventions for sedentary populations. Policy interventions relying solely on tracker distribution may have limited population-level impact without addressing pre-existing motivation and environmental context.

Key words: health trackers, wearables, motivation, behaviour change, qualitative analysis, habit formation

INTRODUCTION

Digital health tracking technologies are changing how individuals around the world manage their health. In Singapore, for example, Healthier SG is increasing the use of trackers by providing them for free with some bonuses for engagement. However, many people who commit to trackers do not necessarily sustain overly long engagement with them; some studies have documented dropout rates higher than 50% in the first six months of use (Patel et al., 2015).

Exploring adults' lived experiences and perceptions of using health trackers will help inform the design of effective digital health interventions. Quantitative studies have examined measures of usage and adoption of health trackers, but qualitative studies will capture the nuances of motivations, frustrations, and shifting interactions users have with the implementation of this technology. Personal experiences, whether it's arriving at a potential goal or the frustration of tracking inaccurate data, affects the perceived value of the tracker, ultimately marking a determinant for sustained use.

The health literature exploring motivation to use a tracker has focused largely on

motivation as a single factor based on theoretical reasoning. For example, self-determination theory posits that intrinsic motivation is more sustainable than externally regulated behaviours (Ryan & Deci, 2000). In contrast, operant conditioning models have suggested that behaviour change may begin with imaginary or other external rewards. Further, while social cognitive theories of motivation highlight social support related to being able to model and share information (Bandura, 1986), the rise of the quantified self-movement suggest that personal tracking may be about curiosity or using data to optimize themselves (Lupton, 2016).

While there is a wide range of motivational perspectives, they often depict motivation as static or poorly defined and a one-dimensional experience. Some newer research has used qualitative inquiry to examine multiple concepts of motivation, highlighting the need for simultaneous multidimensional motivational factors and changes in motivation over time (Nascimento et al., 2018). However, a narrow amount of qualitative inquiry reflects this ongoing shifting, particularly in Asian settings where qualitative research topics are not like Western studies using health trackers.

Research Gaps

First, most research on motivation studies it at a single timepoint in time and does not capture the evolutionary nature of motivation. Second, comparatively little research on motivation considers how different profiles of motivation respond to varying barriers to motivation. Finally, the assumption that the presence of tracking tools will increase activity at the population level remains mostly unproven in terms of qualitative research.

Research Question

What are adults' experiences and perceptions of using health apps and wearables? Specifically, what are the motivations for initial adoption and sustained use, what are the barriers for sustained engagement, and how do adults' experiences and perceptions of health trackers and apps change over time across user profiles?

METHODS

Study Design

The analysis of qualitative themes derived from semi-structured, in-depth interviews. This is a hybrid deductive-inductive method for our analysis: our analysis viewed predetermined four themes identified in the behaviour change literature (extrinsic incentives, gamified goals, social motivation, self-quantification) as a predetermined analysis did take a deductive thematic analysis format, however we maintained an inductive method of analysing themes that emerged through our data (Fereday & Muir-Cochrane, 2006).

Participants

Five adult users of health trackers participated using a purposive sample from Singapore (September-October 2025). Participants varied in age, type of devices used, duration of tracker use, and patterns of health tracker engagement.

Data Collection

Semi-structured interviews lasting 27-38 minutes were conducted online (Microsoft Teams, Zoom) or in-person between September-October 2025.

The interview guide contained questions about:

- Pattern of current health app and wearable device use patterns
- Initial motivations for adoption
- Experiences with daily integration and habit development
- Factors leading to continued use or discontinued use
- Perceived benefits, barriers, and frustrations encountered

- Social influences and peer dynamics
- Future perspectives on long-term use

Interviews were audio-recorded with informed consent and transcribed verbatim for analysis.

Data Analysis

Transcripts were reviewed and analysed iteratively at each stage using Atlas.ti (version 23; ATLAS.ti Scientific Software Development GmbH, Berlin, Germany) through an iterative three-staged analysis process. (see Appendix B, Figures B1-B17 for coding examples)

Stage 1 - Initial Coding - Open coding identified meaningful segments of data, a descriptive label was created, and code segments were grouped into predetermined thematic domains. (Figure B1)

Stage 2 - Theme Refinement - Codes were reviewed for coherence and distinctiveness. Related codes were merged (for example, we merged "routine integration" and "psychological dependence" into "habit formation"). (Figures B2, B5, B10) An unexpected cross-cutting theme emerged: participants' motivations changed from extrinsic to habituated. (Figures B7, B9, B14).

Stage 3 - Pattern Analysis - Code patterns were mapped for relationships among themes, intensities of variations among participants were noted, and contrasting cases were analysed. (Figures B15-B17 show comprehensive dataset views)

Reporting Standards

This study followed the Consolidated Criteria for Reporting Qualitative Research (COREQ) guidelines (Tong et al., 2007) to ensure transparent and comprehensive reporting of qualitative methodology and findings. A complete COREQ checklist is provided in Appendix C.

Ethical Consideration

Participants gave informed consent, were aware of their right to withdrawal, and were provided confidentiality through de-identification.

RESULTS

Results are organized around five main themes, followed by the cross-cutting pattern of motivational evolution.

Experiences and Perceptions

Participants' experiences and perceptions of health tracker use revealed five interconnected themes, each shaping their ongoing engagement. Positive experiences, such as achieving goals and gaining health insights, reinforced continued use, while negative experiences, particularly around accuracy concerns and manual effort requirements, shaped perceptions of tracker utility and influenced discontinuation decisions.

Theme 1: Extrinsic Incentives Drive Initial Adoption

Clearly it was external rewards/bounty that drove initial adoption, and financial incentives were especially impactful. One participant stated: "What motivated me was the vouchers I could redeem. For example, if you hit 5,000 or 10,000 steps, you earn points...So yes, the incentives motivated me more than the design of the tracker" (Kwan). Insurance subsidies also motivated first purchases: "My first device was Fitbit, which I bought because of an insurance-linked incentive with AIA" (Kwan).

The fact that participants had made an investment in devices provided some ongoing motivation as well due to the rationale of sunk costs. As one participant put it: "Another reason is the cost—I paid a lot for this Suunto watch, so I want to maximise its use" (Kwan). Participants observed generational patterns: "Among older adults, like my parents'

generation, many use the free government trackers mainly to earn vouchers" (Kwan).

While extrinsic incentives drove initial adoption, gamification features sustained day-to-day engagement.

Theme 2: Gamified Goals Sustain Daily Engagement

The use of activity rings and progress bars provided strong visual stimuli: "It has what we call activity rings. They set goals for you every day...it motivates you to fill out the bar" (Kathleen). Notifications prompted activity: "sometimes if I sit for too long, my watch will buzz, and it helps me to stand up and move" (Wang).

Users also engaged in compensatory behaviours: "Usually if it's less steps than what I usually do, I try to walk more" (Kathleen). Another noted: "if I haven't taken enough steps that day, I would actually deliberately take some, maybe evening walks" (KF). This goal-responsive behaviour suggests that gamification was effective in translating awareness into behaviour change for willing participants.

Beyond individual goal pursuit, social dynamics emerged as powerful drivers of both adoption and sustained engagement.

Theme 3: Social Motivations Create Interpersonal Accountability

For some, observing peers was the primary adoption trigger: "I had friends who were using this app. They all had Apple Watch, and they would always compare each other's progress...And that was their main motivation to get in this, so I can also do those challenges with my friends" (Kathleen).

Real-time features provided ongoing motivation: "you're actually notified when your friend finishes a workout...I actually feel motivated. I can do maybe better" (Kathleen). Shared use transformed individual exercise into social practice: "Exercising with colleagues or friends makes it fun and motivating...it became like a shared activity" (Kwan).

Device aesthetics had social meaning: "For my age group, it's different—it's more about status. If you're wearing the free tracker, it doesn't look as 'cool' compared to an Apple Watch" (Kwan). However, social features also contributed to discomfort: "So for those who I'm not very close with...I try to hide my activity from them" (Kathleen).

Importantly, some explicitly denied social influence: "Neither am I doing it because everyone else is. It's just purely out of my own personal curiosity" (KF).

This individual curiosity distinct from external rewards or social pressure, reflected a fourth motivational domain centred on data and self-knowledge.

Theme 4: Self-Quantification Satisfies Curiosity and Enables Decisions

Initial curiosity drove exploration: "So it was the curiosity that makes me wonder...what actually happens during my sleep...how much of deep sleep, REM sleep" (KF). One user expressed: "I want to have a more quantitative measurement of my body" (ZL).

Athletes leverage data to optimise: "Knowing the stats makes me more conscious...If I see that I only spent 5 minutes at high intensity, I'll tell myself that next time I want to push harder" (Kwan). Self-described "number people" valued quantification intrinsically: "Well, I'm a quantitative guy. So to me everything that if I can do more quantifiable...I like it" (ZL).

Perhaps most powerfully, numbers prevent self-deception: "for somebody who likes and trusts numbers...These numbers give me clear visibility...to get me away from that 'I think' mindset to 'I know'...in the past without these numbers, I kind of convinced myself, 'I've walked enough,' but today, the number tells me exactly...there's no way...no running away from that number" (KF).

Sleep tracking filled gaps of information valued by the individual: "My sleep has been quite poor recently, but the watch shows my sleep duration and patterns...It also made me more aware...it sometimes tells me I've over-exercised and need to rest" (Kwan).

Regardless of a strong motivation in these four options, all participants faced some barriers that impacted sustained use, even if their tolerance of those barriers depended on the

type of barrier and/or user profile.

Theme 5: Barriers to Sustained Use - Differential Tolerance by Type

For both discontinued use (in particular, nutrition apps) and indignation with wearables, we could observe critical patterns of barriers that influenced whether users continued or stopped using trackers.

Manual Effort Requirements Led to Abandonment

The manual entry requirements for tracking nutrition created an active barrier that caused users to stop, even highly motivated users: "Firstly, it was troublesome to log every single food that I have eaten...most of these apps...are pretty simple recipes or single ingredient foods...if I...eat Buddha jumping over the wall [complex Asian dish], you can never find this...Even for hard-boiled egg...there were at least three or four options with different calorie counts...what's the difference? A hard-boiled egg is a hard-boiled egg...My experience back then has not been a very motivating one" (KF). This detailed account revealed multiple compounding barriers: Effort burden, lack of database (cultural/contextual appropriateness), and clarity of data reducing credibility. However, this combination of an active effort requirement and an uncertainty of credibility resulted in complete abandonment of the nutrition tracker.

Technical Limitations Frustrated but Didn't End Use

Although the shortcomings of the devices frustrated the participants, they were still used by motivated participants: "Xiaomi tracker can auto-detect, but with this watch I have to manually start recording. If I forget to press start, nothing gets recorded, which is frustrating...when I do HIIT, there isn't a HIIT mode—I just must select something like 'aerobics.' When I play pickleball, there's no option, so I just use 'racket sports.'...It's strange because the watch has modes like skiing and snowboarding, but no pickleball or HIIT, which are more common here" (Kwan). Although the limitations created user annoyance, this annoyance was tolerated, suggesting the passive barriers (one-time or occasional issues) differ from active barriers (ongoing effort) in terms of risk of discontinuation.

Accuracy Concerns Tolerated Despite Doubts

Users continued despite accuracy skepticism: "Another issue is the accuracy of the data, since different brands show different results...Even when exercising with my colleagues one using a Samsung watch, another using Fitbit, and another with Apple Watch we got very different calorie and heart rate readings...My main expectation now is accuracy" (Kwan). Yet another participant-maintained belief despite peer skepticism: "there have been a lot of comments...from my friends that...the accuracy are questionable, right and but to me is, well, all these are hearsay, so I still think that...all these trackers, the sensors...has a certain level of accuracy to it" (KF). This "good enough" tolerance suggested users accepted imperfection if utility threshold was met. The pattern that emerged was: active barriers requiring an ongoing effort, resulted in the users abandoning it, while passive barriers, users who were more motivated, therefore would tolerate the inaccurate readings for all the other utility the product offers.

Beyond the motivational domains and barriers related to motivation, a significant pattern around time emerged across all user types: motivations were not necessarily constant and did evolve through different stages.

Cross-Cutting Pattern: Motivational Evolution and Habit Formation

Motivation Development and Habit Creation The participants arrived using various pathways outside incentives, such as curiosity, social influence, or convenience. Eventually, passive observing developed into active responding: "It has definitely changed over the years...When I first started, it was just, 'good to know that I walk say a thousand steps'...But today, I look at it and I tell myself, 'I need to take more steps today'" (KF).

The stated participants established a habit of regular use: "Yes, definitely. Now I feel strange if I'm not wearing a watch. It has become a habit. For example, when I swim, I

automatically put on the watch" (Kwan). Another noted: "it's more of already part of my routine...I feel weird when I don't open the app" (Kathleen). The language of "strange," "weird," and "automatically" signalled unconscious behavioural integration.

Unexpected Finding: Trackers Enhance Existing Motivation Rather Than Create It

Several of the participants described independently experiencing: "most of the people that I know that has started using fitness band, they have been pretty active to begin with...they have already been doing it" (KF). Another stated: "I am quite active. I do love working out and that's why I got the app to begin with" (Kathleen).

Most explicitly: "if someone wants to lose weight, a tracker is very useful...For those very motivated in sports, like training for a marathon, the stats are crucial...But if someone isn't motivated to exercise, then even with a tracker, it probably won't help. So, I think motivation is the key factor" (KF).

One participant described environmental limitations: "whether those nudges will work depends on your environment...if you're just at home and you don't have a space to exercise...you really are not in the space which allows you to work out...besides the app, shaping the environment would also be helpful" (Kathleen).

Even among data-conscious users, numbers didn't universally change behaviour: "I do know people who looks at the number and 'yeah, okay, it's 2000...it's 8000'" (KF).

Discussion

This investigation reveals four motivational domains extrinsic rewards, gamified objectives, social motivation, self-quantification, and one critical implementation factor barriers to sustained use, with a key chronological element: motivations shifted from extrinsic motivators to ingrained behaviours. The key takeaway is that trackers do not create new motivation; rather, pre-existing motivation influences how trackers are used, depending on profile type, barrier type, and context.

Adult user experiences and perceptions of health trackers were highly individualised and fluid. Positive experiences (e.g., "the number tells me exactly whether I have clocked enough steps") informed how the participants perceived the value of the tracker, while negative experiences (e.g., "I find it bothersome to log every single food") lowered perceived utility. Perceptions were also not fixed; users talked about progressing from scepticism to habitual reliance or going from motivation to frustration to giving up. This suggests longitudinal rather than snapshot assessments are needed.

Multi-Dimensional Motivation

The evidence that users exhibited multiple, simultaneous motivations was consistent with other recent research that shows that engagement in trackers is more multi-faceted than other single theory models of motivation (Nascimento et al., 2018). The data extended the evidence of multi-faceted motivation by documenting how the relative importance of each domain changed over time and to address the question of when each of these motivations was activated. We found that extrinsic motivation facilitates initial adoption, gamification is effective for sustaining daily use, social had a competitive/civic dimension, and self-quantification reflected an interest in analytics and data. The layering sequence has implications that contradict theories of motivation hierarchy putting intrinsic motivation above all others (Deci & Ryan, 2000) and suggests that using a pluralistic motivation model may be better for activating greater and sustained user engagement.

Motivational Evolution

The documented progression from conscious motivation to automatic habit represents critical insight often missing from cross-sectional studies. According to habit formation literature, behaviours that occur in repetition in a stable context become activated by a cue and require little conscious intent (Lally et al., 2010). Much of the data in this study support this psychological mechanism in the habit formation process - participants described using

friends, family, or acquaintances as "weird" to omit and "automatically" performed. This process of evolution from conscious desire to automatic response, indicates interventions should create a sequencing of strategy: first use extrinsic incentives to adopt a behaviours, then use gamification elements to promote engagement, and finally design for habit formation to promote sustainability.

The "Already Active" Problem

Perhaps most consequential: trackers enhance existing motivation rather than creating it de novo. This challenges assumptions behind efforts aimed broadly at distributing trackers for population levels. Our participants explicitly stated their concern that trackers assist rather than motivate individuals, and they acknowledged they help motivated individuals with optimisation instead of generating motivation in previously unmotivated populations. This aligns with Rogers Diffusion of Innovations theory (2003) which posits that early adopters should differ systematically from late adopters; in this case, that the early adopters were already health conscious. This too resonates with a randomised controlled trial (RCT) that provided low evidence of effect on sedentary adults unless other factors supported the efforts (Brickwood et al., 2019). This suggests that the effectiveness of trackers in the deployment stage may come down to Rothman's (2000) concepts of initiation and maintenance of behaviour change. Specifically, that trackers may assist with maintenance but will not initiate behaviour changes. The implications for public policy are significant; a free to the population tracker may have limited impact on population health without an intervention that addresses motivation to change behaviour and the introduction of the environmental habit context.

Barrier Typology

The distinction between active barriers (requiring sustained effort and engagement) and passive barriers (only a problem sometimes) is fruitful in design consideration. Nutrition tracking failed despite being a strong motivator, because it required continuous, manual logging and was a cultural mismatch for students. Conversely, wearable activity tracking succeeded despite the possibility of inaccuracies, because it took the measurement burden off students. This observation reflects Fogg's (2009) Behaviour Model which emphasises that "simplicity" is a key behavioural feature. Our data specify what simplicity means in the context of wearables: automation is better than comprehensiveness, and as passive (or minimal effort required). It is important to note that these passive (or minimal-time and effort required) measures could have been combined with motivational and environmental interventions.

Barriers and User Tolerance

It is noteworthy that barriers emerged as a unique theme, something beyond merely a barrier to motivation. The important difference between active and passive barriers explained abandonment behaviour better than motivation itself. This extends Fogg's (2009) Behaviour Model by clarifying that "ability" means reducing active barriers (i.e., effort over time), rather than making it perfect. In nutrition tracking, we see the abandonment despite high motivation at the start. This means high motivated users have threshold limits on tolerating the sustained cognitive and time-based effort put in to address barriers. Conversely, the tolerance for wearables' data inaccuracies points to a "satisficing" model (Simon, 1956) while still allowing participants to develop a behaviour whereby "good enough" data could be implemented when the effort-to-benefit ratio is favourable. This is relevant for intervention design; intervention design should design methods that rely on passive data collection (automated tracking) instead of intense effort (manual logging) where although the latter might provide more comprehensive data and accuracy. The cultural mismatch with Asian food databases also highlights a call for localized content for global health technologies.

User Profiles

Distinct profiles emerged: data-driven optimisers, social fitness enthusiasts, incentive-responsive adopters, convenience-first users, suggesting one-size-fits-all approaches are

suboptimal. This is consistent with the argument of Personalisation Theory that more personalised components enhance the effectiveness of interventions as opposed to generic interventions (Dijkstra, 2008). Practically, this implies that components for engagement should be developed in clear segments: for users driven by quantitative data, it is important to focus on accuracy; for users who are social, prioritise the opportunity to connect with friends; for users who are incentivised adopters, keep reward but have transition pathways for independence; for users who are convenience use, focus on integration into multipurpose devices.

Strengths and Limitations

Strengths: Detailed qualitative investigation of motivational change; comprehensive and hybrid methodology effectively balanced a structured theoretical framework while permitting unexpected patterns to be emerging; rich quotations support consideration of interpretive validity; sample consisted of diverse representatives which demonstrates heterogeneity.

Limitations: Small sample (n=5) limits generalisability, though thematic saturation was achieved. Discontinued users were underrepresented. Recall bias may limit the accuracy of retrospectively recalling earlier incentive programme experiences. Selection bias may have been involved as those willing to participate may have been more engaged and less detached in the intervention than one participant who declined. Finally, the Singapore context's incentive programmes may not be universal to other contexts. The analysis was limited to motivation and did not assess programme health outcomes.

Implications

For public health interventions: Couple tracker delivery with motivational interviewing and graduated support and positioning trackers as maintenance tools with realistic expectations.

For tracker design: Value automation over comprehensiveness, "good enough" accuracy over perfect measurement, social features that can be configured, cultural localisation, and habit formation.

For healthcare integration: Consider integration possibilities while recognising the challenges to accuracy, interoperability and privacy.

CONCLUSION

Motivation for using health trackers is multidimensional, evolves over time, and is highly individualised. Although extrinsic motivators, gamified targets, social aspects, and self-quantification may all play a role, the significance of these factors is contextual to profile and shifts over time, to eventually become habitual. Critically, the difference between active barriers that needed to be overcome with effort on an on-going basis, and passive barriers that meant only facing a problem every now and again, was a significant factor for all users and nearly all of the metrics for sustained use, as users would drop trackers that had high active barriers whether they were motivated or not. Importantly, and in contradiction of the research, our findings question the assumption that providing health trackers will increase activity levels for populations. Health trackers may support pre-existing motivations, rather than create motivations, acting purely as tools for optimise for the individual who is already physically active but not as interventions for physical activity change of a sedentary population. This underlying "already active" behaviour, and the different levels of personal toleration for barriers, and varying degrees of necessity at the environmental level suggests that interfacing technology for intervention require supplementary means of addressing both the fundamental motivations, environmental constraints and technology interventions. In practice, we suggest that health tracker programming may be part of a broad and targeted intervention. For designers, make tracking the default and allow cultural appropriateness for identities and using configurable social elements. For researchers, embrace long-term, culturally situated, longitudinal programme studies to begin to map the connections between

motivational pathways toward outcomes. As with messaging on health trackers, the various features of a health trackers are not silver bullet solutions, rather successes depend upon the match between user profile with features, sequences of motivation strategies, and limitations of technologies as interventions. As one participant noted, "motivation is the key factor"—technology may be helpful in amplifying motivation —seldom in creating motivation

REFERENCES

ATLAS.ti Scientific Software Development GmbH. (2023). *ATLAS.ti* (Version 23) [Computer software]. https://atlasti.com

Allison Tong, Peter Sainsbury, Jonathan Craig, Consolidated criteria for reporting qualitative research (COREQ): a 32-item checklist for interviews and focus groups, *International Journal for Quality in Health Care*, Volume 19, Issue 6, December 2007, Pages 349–357, https://doi.org/10.1093/intqhc/mzm042

Bandura, A., & National Inst of Mental Health. (1986). *Social foundations of thought and action*: *A social cognitive theory*. Prentice-Hall, Inc.

Brickwood, K. J., Watson, G., O'Brien, J., & Williams, A. D. Consumer-Based Wearable Activity Trackers Increase Physical Activity Participation: Systematic Review and Meta-Analysis. JMIR Mhealth Uhealth. 2019 Apr 12;7(4):e11819. doi: 10.2196/11819. PMID: 30977740; PMCID: PMC6484266.Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behaviour. *Psychological Inquiry, 11*(4), 227-268.

Dijkstra, A. (2008). The psychology of tailoring-ingredients in computer-tailored persuasion. Social and Personality Psychology Compass, 2(2), 765–784. https://doi.org/10.1111/j.1751-9004.2008.00081.x.

Fereday, J., & Muir-Cochrane, E. (2006). Demonstrating Rigor Using Thematic Analysis: A Hybrid Approach of Inductive and Deductive Coding and Theme Development. International Journal of Qualitative Methods, 5(1), 80-92. https://doi.org/10.1177/160940690600500107 (Original work published 2006)

Fogg, B. J. (2009). A behaviour model for persuasive design. *Proceedings of the 4th International Conference on Persuasive Technology*, Article 40.

Lally, P., van Jaarsveld, C. H. M., Potts, H. W. W., & Wardle, J. (2010). How are habits formed: Modelling habit formation in the real world. European Journal of Social Psychology, 40(6), 998–1009. https://doi.org/10.1002/ejsp.674

Lupton, D. (2016). *The quantified self*. Polity Press.

Nascimento, B., Oliveira, T., & Tam, C. (2018). Wearable technology: What explains continuance intention in smartwatches? *Journal of Retailing and Consumer Services, 43*, 157-169. https://doi.org/10.1016/j.jretconser.2018.03.017

Patel MS, Asch DA, Volpp KG. Wearable devices as facilitators, not drivers, of health behaviour change. JAMA. 2015 Feb 3;313(5):459-60. doi: 10.1001/jama.2014.14781. PMID: 25569175.

Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.

Rothman AJ. Toward a theory-based analysis of behavioural maintenance. Health Psychol. 2000 Jan;19(1S):64-9. doi: 10.1037/0278-6133.19.suppl1.64. PMID: 10709949.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. American Psychologist, 55(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

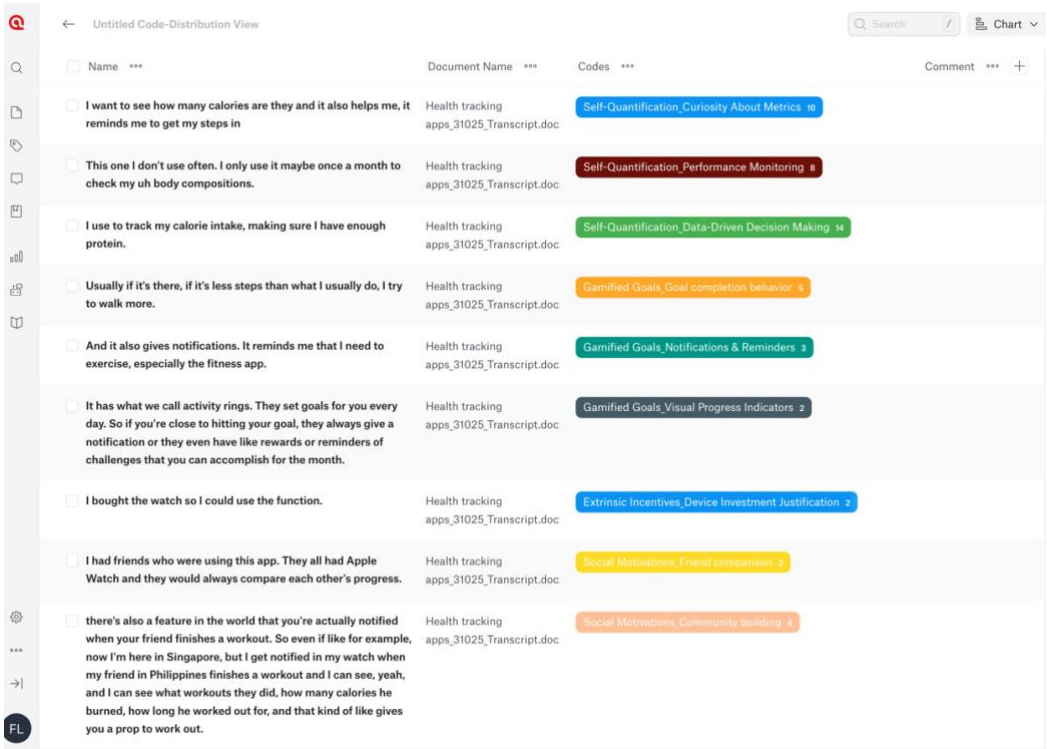
Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review, 63*(2), 129–138. https://doi.org/10.1037/h0042769

APPENDIX A: CODING STRUCTURE

Category	Sub-Themes
Extrinsic Incentives	Financial Rewards, Insurance Benefits, Device Investment, Cost Concerns
Gamified Goals	Visual Progress Indicators, Notifications & Reminders, Goal Completion Behaviour
Social Motivations	Friend Comparison, Community Building, Communal Activity, Social Image & Status, Privacy Concerns, Peer Observation, Lack of Social Influence
Self-Quantification	Curiosity About Metrics, Performance Monitoring, Data-Driven Decision Making, Accountability Function, Sleep & Recovery Tracking, Trend Analysis
Motivational Evolution	Stage Progression, Automatic Behaviour, Habit Formation, Sustained Use, Individual Variation
Barriers	Manual Effort Required, Accuracy Concerns, Technical Limitations

Figure 1: Coding Structure

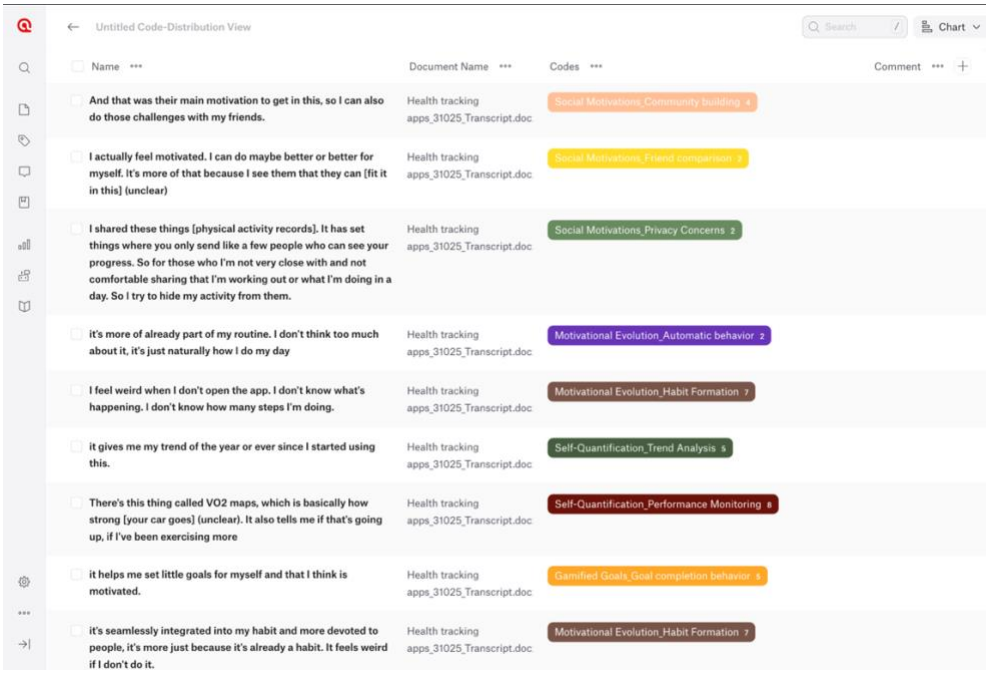
APPENDIX B: CODEBOOK FROM ATLAS.TI



Untitled Code-Distribution View

Name	Document Name	Codes	Comment
I want to see how many calories are they and it also helps me, it reminds me to get my steps in	Health tracking apps_31025_Transcript.doc	Self-Quantification_Curiosity About Metrics 10	
This one I don't use often. I only use it maybe once a month to check my uh body compositions.	Health tracking apps_31025_Transcript.doc	Self-Quantification_Performance Monitoring 8	
I use to track my calorie intake, making sure I have enough protein.	Health tracking apps_31025_Transcript.doc	Self-Quantification_Data-Driven Decision Making 14	
Usually if it's there, if it's less steps than what I usually do, I try to walk more.	Health tracking apps_31025_Transcript.doc	Gamified Goals_Goal completion behavior 5	
And it also gives notifications. It reminds me that I need to exercise, especially the fitness app.	Health tracking apps_31025_Transcript.doc	Gamified Goals_Notifications & Reminders 3	
It has what we call activity rings. They set goals for you every day. So if you're close to hitting your goal, they always give a notification or they even have like rewards or reminders of challenges that you can accomplish for the month.	Health tracking apps_31025_Transcript.doc	Gamified Goals_Visual Progress Indicators 2	
I bought the watch so I could use the function.	Health tracking apps_31025_Transcript.doc	Extrinsic Incentives_Device Investment Justification 2	
I had friends who were using this app. They all had Apple Watch and they would always compare each other's progress.	Health tracking apps_31025_Transcript.doc	Social Motivations_Friend comparison 2	
there's also a feature in the world that you're actually notified when your friend finishes a workout. So even if like for example, now I'm here in Singapore, but I get notified in my watch when my friend in Philippines finishes a workout and I can see, yeah, and I can see what workouts they did, how many calories he burned, how long he worked out for, and that kind of like gives you a prop to work out.	Health tracking apps_31025_Transcript.doc	Social Motivations_Community building 4	

Figure B1: Figure B1: Sample Coding - Self-Quantification, Gamified Goals, Extrinsic Incentives, and Social Motivations. Numbers indicate frequency of code application across the dataset.



Untitled Code-Distribution View

Name	Document Name	Codes	Comment
And that was their main motivation to get in this, so I can also do those challenges with my friends.	Health tracking apps_31025_Transcript.doc	Social Motivations_Community building 4	
I actually feel motivated. I can do maybe better or better for myself. It's more of that because I see them that they can [fit it in this] (unclear)	Health tracking apps_31025_Transcript.doc	Social Motivations_Friend comparison 2	
I shared these things [physical activity records]. It has set things where you only send like a few people who can see your progress. So for those who I'm not very close with and not comfortable sharing that I'm working out or what I'm doing in a day. So I try to hide my activity from them.	Health tracking apps_31025_Transcript.doc	Social Motivations_Privacy Concerns 2	
it's more of already part of my routine. I don't think too much about it, it's just naturally how I do my day	Health tracking apps_31025_Transcript.doc	Motivational Evolution_Automatic behavior 2	
I feel weird when I don't open the app. I don't know what's happening. I don't know how many steps I'm doing.	Health tracking apps_31025_Transcript.doc	Motivational Evolution_Habit Formation 7	
it gives me my trend of the year or ever since I started using this.	Health tracking apps_31025_Transcript.doc	Self-Quantification_Trend Analysis 5	
There's this thing called VO2 maps, which is basically how strong [your car goes] (unclear). It also tells me if that's going up, if I've been exercising more	Health tracking apps_31025_Transcript.doc	Self-Quantification_Performance Monitoring 8	
it helps me set little goals for myself and that I think is motivated.	Health tracking apps_31025_Transcript.doc	Gamified Goals_Goal completion behavior 5	
it's seamlessly integrated into my habit and more devoted to people, it's more just because it's already a habit. It feels weird if I don't do it.	Health tracking apps_31025_Transcript.doc	Motivational Evolution_Habit Formation 7	

Figure B2: Sample Coding - Self-Quantification, Gamified Goals, Motivational Evolution, and Social Motivations. Numbers indicate frequency of code application across the dataset.

Lifestyle is a form of prescription too—exercise and sleep management can help treat or manage disease	Interview trascript .docx	Motivational Evolution_Habit Formation 7
If trackers can feed data into the system, it could really support healthcare, like how the Libre system helps diabetes patients track sugar levels. But Libre doesn't show exercise data—so combining these would be powerful.	Interview trascript .docx	Self-Quantification_Data-Driven Decision Making 14
n experiences, what advice would you give to someone your age who is considering starting to use a health tracker?	Interview trascript .docx	
I'd say it depends on the person's health goals and financial situation	Interview trascript .docx	Motivational Evolution_individual variation 1
For example, if someone wants to lose weight, a tracker is very useful.	Interview trascript .docx	Self-Quantification_Trend Analysis 5
Or if they have a health condition—like a colleague of mine with irregular heart rhythms—the watch can alert him if his heart rate goes too high during exercise, which is important for safety.	Interview trascript .docx	Social Motivations_Peer observation 3
For those who are very motivated in sports, like training for a marathon, the stats are crucial. I have a friend and even my boss who are serious runners—they use the watch to pace themselves and track training. For them, it's essential.	Interview trascript .docx	Self-Quantification_Performance Monitoring 8
But if someone isn't motivated to exercise, then even with a tracker, it probably won't help. So I think motivation is the key factor.	Interview trascript .docx	Barriers_Technical limitations 4

Figure B3: Sample Coding - Self-Quantification, Barriers, Motivational Evolution, and Social Motivations. Numbers indicate frequency of code application across the dataset.

Another issue is the accuracy of the data, since different brands show different results.	Interview trascript .docx	Barriers_Accuracy concerns 3
when I do HIIT, there isn't a HIIT mode—I just have to select something like "aerobics."	Interview trascript .docx	Barriers_Technical limitations 4
When I play pickleball, there's no option, so I just use "racket sports."	Interview trascript .docx	Barriers_Technical limitations 4
It's strange because the watch has modes like skiing and snowboarding, but no pickleball or HIIT, which are more common here.	Interview trascript .docx	Barriers_Technical limitations 4
if I grow old and develop dementia, it would be helpful if the watch had GPS tracking so my family could locate me	Interview trascript .docx	Social Motivaitons_Social Responsibility 1
I actually know colleagues who bought GPS watches for their children, to keep track of where they are. I think that's also useful for elderly people.	Interview trascript .docx	Social Motivations_Peer observation 3
For checking my health, I think it depends on accuracy. If the data is reliable, then yes, I would rely on it more.	Interview trascript .docx	Self-Quantification_Data-Driven Decision Making 14
Ideally, exercise and lifestyle data could be integrated into the healthcare system. For example, doctors could use it alongside medical records, like blood tests, to get a more holistic view of a patient's health.	Interview trascript .docx	Motivational Evolution_Habit Formation 7
Lifestyle is a form of prescription too—exercise and sleep management can help treat or manage disease	Interview trascript .docx	Motivational Evolution_Habit Formation 7

Figure B4: Sample Coding - Self-Quantification, Barriers, Motivational Evolution, and Social Motivations. Numbers indicate frequency of code application across the dataset.

Untitled Code-Distribution View

Name	Document Name	Codes	Comment
I think it's really just a matter of doing consistence because if you do it enough times, then it feels natural to you	Health tracking apps_31025_Transcript.doc	Motivational Evolution_Habit Formation 7	
I think what also helped me make it a habit is because my friends were also doing it. And so we do it together and just having that community, having that same in the mindset of the people, it's up to you so I'm I think.	Health tracking apps_31025_Transcript.doc	Social Motivations_Communal Activity 2	
having the numbers, I burned this amount of calories. I think it kind of like tricks your brain for thinking if you've been working hard on it or not. So yeah, I think that's also one thing. Because it's measurable. You know, maybe that's a better way of putting it. It becomes measurable.	Health tracking apps_31025_Transcript.doc	Self-Quantification_Data-Driven Decision Making 14	
I think naturally when you're doing it with other people, it's easier to stay. That's if you feel motivated, that other person can actually push you to do it,	Health tracking apps_31025_Transcript.doc	Social Motivations_Community building 4	
I am quite active. I do love working out and that's why I got the app to begin with	Health tracking apps_31025_Transcript.doc		
I see my progress in the more measurable way because it shows the family spread and like how many days I've been doing it, how many times.	Health tracking apps_31025_Transcript.doc	Self-Quantification_Trend Analysis 5	
nitially, it was a plus, because at first I just wanted the watch to, you know, tap in and out of transport. But then this watch can also track health, so I just used it to track my health. Because I might as well know how many steps you're walking.	IDI on the Use of Digital Health Apps_Transcript.docx	Self-Quantification_Curiosity About Metrics 10	
It has a like, I don't know what it has, it has like a power bar, like you fill out the bar and then it's off. You fill out the bar, and then it's like it will, it will show you how much progress you've made for the day. So I just think it's a bit more helpful because it motivates you to fill out the bar.	IDI on the Use of Digital Health Apps_Transcript.docx	Gamified Goals_Visual Progress Indicators 2	

Figure B5: Sample Coding - Self-Quantification, Gamified Goals, Motivational Evolution, and Social Motivations. Numbers indicate frequency of code application across the dataset.

I think accuracy, better accuracy, because I'm quite sure this watch is not that accurate.	IDI on the Use of Digital Health Apps_Transcript.docx	Barriers_Accuracy concerns 3
It tracks the training of your arms and steps. It tracks your movement and steps. So sometimes I can be sitting down and my steps will be going up	IDI on the Use of Digital Health Apps_Transcript.docx	Self-Quantification_Data-Driven Decision Making 14
I think, after using two years, it's, like, a habit to wear away with it	IDI on the Use of Digital Health Apps_Transcript.docx	Motivational Evolution_Habit Formation 7
I mean, like when I go to the gym, it's easier to track what's my heart rate and all that. So I can, you know like, a certain amount of, within a certain range itself is better for fat burning and all that.	IDI on the Use of Digital Health Apps_Transcript.docx	Self-Quantification_Curiosity About Metrics 10
sometimes if I sit for too long, my watch will bust, and it helps me to stand up and move around a bit	IDI on the Use of Digital Health Apps_Transcript.docx	Gamified Goals_Sedentary alerts 1
Because it's convenient when you have a number to hear. Because right now everybody is talking about the 10,000 steps and all that. So then it becomes like your device can tell you... Yeah, yeah, exactly how many steps have you done	IDI on the Use of Digital Health Apps_Transcript.docx	Self-Quantification_Data-Driven Decision Making 14
es, yes, I think I will wear this long-term. I will change devices, but I will wear devices quite often.	IDI on the Use of Digital Health Apps_Transcript.docx	Motivational Evolution_Sustained use 3
my first use of the wearable watches, the main purpose, just to watch my heartbeat when I exercise.	Transcript-ZL.docx	Self-Quantification_Curiosity About Metrics 10

Figure B6: Sample Coding - Self-Quantification, Gamified Goals, Motivational Evolution, Social Motivations, and Barriers. Numbers indicate frequency of code application across the dataset.

So I want to know, I want to have a more quantitative measurement of my body.	Transcript-ZL.docx	Self-Quantification_Curiosity About Metrics 10
as you get, get more in into working out and you wanna measure your, the consequences of your, your working out	Transcript-ZL.docx	Self-Quantification_Performance Monitoring 8
I'd say the accuracy. Well, I'll definitely just wanna know the, the accurate heartbeat. OK, so it needs to be accurate.	Transcript-ZL.docx	Self-Quantification_Curiosity About Metrics 10
So after a while after you using, the, using them for the while, then I start to, well, you start to get a sense of what your heartbeat will be	Transcript-ZL.docx	Motivational Evolution_Reduced dependence 1
You know, you know, you know 100 grammes of, of this is worth how much calorie, then you start to use them less. So you memorise this.	Transcript-ZL.docx	Motivational Evolution_Knowledge internalization 1
Well, I'm a quantitative guy. So to me everything that if I can do more quantifiable, or, in a more quantifiable way, I like it.	Transcript-ZL.docx	Self-Quantification_Data-Driven Decision Making 14
So with those apps, I can quantify my, eh, like the food intake, the calories, I don't know, I can qualify my, my workout effect, so it helps.	Transcript-ZL.docx	Self-Quantification_Data-Driven Decision Making 14
So if you continue work out, you wanna know, OK, I have, I mean, you will continue on to know the effect and plus as time goes by these products they upgrade, they have more functions than they measure.	Transcript-ZL.docx	Motivational Evolution_Expanding curiosity 1
Or if one day they start to charge a very high price. OK, so far they are all free	Transcript-ZL.docx	Extrinsic Incentives_Financial Concerns 1
At one point, if you feel that it's not worth it, then I'm I might consider stop using it.	Transcript-ZL.docx	Extrinsic Incentives_Value Assessment 1

Figure B7: Sample Coding - Self-Quantification, Extrinsic Incentives, and Motivational Evolution. Numbers indicate frequency of code application across the dataset.

I guess the main limitation is to how professional they are. I mean there are just too many of them and then you don't know, I mean many of them are not really professional health equipment producers. They're just, they just created this product and let you feel right. But, but I think the more professional it is, the better it is.	Transcript-ZL.docx	Self-Quantification_Data-Driven Decision Making 14
They can measure more of, your body's situations. Well it's just as what I mentioned. When I first started, It's only heartbeat. It's only the body mass, the fat mass, the weight, the muscle mass. But later on, I mean can just move on to many many other aspects of your body, like blood pressure, blood sugar level and then many other things. I can't. I can't name all, like how well you sleep and your pressure level, I mean mental stress levels, all those stuff, right?	Transcript-ZL.docx	Self-Quantification_Trend Analysis 5
If you buy it and it does not tell you anything. It's like garbage-in garbage-out. What do you want it?	Transcript-ZL.docx	Self-Quantification_Curiosity About Metrics 10
You can turn them off. It's not. It's never OK, it's not (a problem).	Transcript-ZL.docx	Gamified Goals_Notifications & Reminders 3
It tracks about 99 different activities. I sync it to my phone so I can see all the records. For example, when I swam yesterday, it showed me the distance, time, heart rate, and even sleep data.	Interview trascript .docx	Self-Quantification_Performance Monitoring 8
My sleep has been quite poor recently, but the watch shows my sleep duration and patterns	Interview trascript .docx	Self-Quantification_Sleep & Recovery Tracking 1
or example, if I sit for two hours, it tells me to move.	Interview trascript .docx	Gamified Goals_Notifications & Reminders 3
No, I don't share my data. I prefer to keep it private.	Interview trascript .docx	Social Motivations_Privacy Concerns 2

Figure B8: Sample Coding - Self-Quantification, Gamified Goals, and Social Motivational. Numbers indicate frequency of code application across the dataset.

But honestly, I didn't use it because it was free—it was mainly because of the incentives. The watch itself wasn't attractive. Compared to my friends' Apple Watches, the free one looked very basic. What motivated me was the vouchers I could redeem.	Interview trascript .docx	Extrinsic Incentives_Financial Rewards 4
For example, if you hit 5,000 or 10,000 steps, you earn points. You can also scan QR codes at supermarkets or food courts to earn more. 750 points gave you $5, 1500 points gave you $10.	Interview trascript .docx	Extrinsic Incentives_Financial Rewards 4
So yes, the incentives motivated me more than the design of the tracker.	Interview trascript .docx	Extrinsic Incentives_Financial Rewards 4
At first, honestly, I didn't have much expectation. But now, I think sleep tracking matters a lot to me because my sleep has been poor.	Interview trascript .docx	Motivational Evolution_Stage progression 4
Another reason is the cost—I paid a lot for this Suunto watch, so I want to maximize its use	Interview trascript .docx	Extrinsic Incentives_Device Investment Justification 2
My main expectation now is accuracy.	Interview trascript .docx	Self-Quantification_Curiosity About Metrics 10
Even when exercising with my colleagues—one using a Samsung watch, another using Fitbit, and another with Apple Watch—we got very different calorie and heart rate readings.	Interview trascript .docx	Self-Quantification_Curiosity About Metrics 10
Yes, definitely. Now I feel strange if I'm not wearing a watch. It has become a habit.	Interview trascript .docx	Motivational Evolution_Habit Formation 7
For example, when I swim, I automatically put on the watch to track my stroke rate and distance.	Interview trascript .docx	Motivational Evolution_Automatic behavior 2

Figure B9: Sample Coding - Self-Quantification, Extrinsic Incentives, and Motivational Evolution. Numbers indicate frequency of code application across the dataset.

or example, when I swim, I automatically put on the watch to track my stroke rate and distance.	Interview trascript .docx	Motivational Evolution_Automatic behavior 3
Over time, it has motivated me to exercise more and track my activities.	Interview trascript .docx	Motivational Evolution_Sustained use 3
It also made me more aware of my sleep. For example, it sometimes tells me I've over-exercised and need to rest	Interview trascript .docx	Self-Quantification_Data-Driven Decision Making 14
I also set sleep targets—like 8 hours—but I usually fail to meet them. When I wake up and see the sleep stats, I often feel disappointed. But it also nudges me to try sleeping earlier next time.	Interview trascript .docx	Gamified Goals_Goal completion behavior 5
Exercising with colleagues or friends makes it fun and motivating.	Interview trascript .docx	Social Motivations_Community building 4
For example, one of my colleagues has an Apple Watch. At first, she would start her workout and record it on the watch, and I would just watch her do it. After some time, I started recording my workouts too—it became like a shared activity.	Interview trascript .docx	Social Motivations_Communal Activity 2
Knowing the stats makes me more conscious. For instance, the watch shows my workout intensity based on my heart rate.	Interview trascript .docx	Self-Quantification_Data-Driven Decision Making 14
If I see that I only spent 5 minutes at high intensity, I'll tell myself that next time I want to push harder and spend more time at the higher level.	Interview trascript .docx	Self-Quantification_Performance Monitoring 8
Xiaomi tracker can auto-detect, but with this watch I have to manually start recording. If I forget to press start, nothing gets recorded, which is frustrating.	Interview trascript .docx	Barriers_Manual effort required 1

Figure B10: Sample Coding - Self-Quantification, Gamified Goals, Motivational Evolution, Social Motivations, and Barriers. Numbers indicate frequency of code application across the dataset.

My first device was Fitbit, which I bought because of an insurance-linked incentive with AIA—it was discounted if you bought it with the plan.	Interview trascript .docx	Extrinsic incentives_Insurance 1
Among older adults, like my parents' generation, many use the free government trackers mainly to earn vouchers.	Interview trascript .docx	Extrinsic Incentives_Financial Rewards 4
For my age group, it's different—it's more about status	Interview trascript .docx	Social Motivations_Social Image & Status 2
If you're wearing the free tracker, it doesn't look as "cool" compared to an Apple Watch or other branded watches.	Interview trascript .docx	Social Motivations_Social Image & Status 2
on days where a two hour session um... gives me a reading of more than a thousand two (calories burnt) , I know that it has been a more intense game and the opposite applies.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Performance Monitoring 8
So it so it was it was the curiosity that makes me wonder, hey, you know, what actually happens during my sleep, right? And err... how much of deep sleep REM (Rapid Eye movement) sleep and etc. that I go through during the entire sleep cycle	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Curiosity About Metrics 10
Neither am I um... doing it because everyone else is. It's just purely out of my own personal curiosity	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Social Motivations_Lack of social influence 1

Figure B11: Sample Coding - Self-Quantification, Extrinsic Incentives, and Social Motivations. Numbers indicate frequency of code application across the dataset.

for somebody who likes and trust numbers, right? These numbers give me clear visibility and data for myself to know that err... to get me away from that "I think" mindset to "I know that I have walked X number of steps a day	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Data-Driven Decision Making 14
t has definitely changed over the years, right? Um... When I first started, it was just, "Eh, good to know that I walk like say a thousand steps that day" Right? And you know maybe 8 thousand steps another day. But I don't do anything about it, but I hit a thousand (steps), 8,000 (steps), 10 thousand (steps), right? But at this point of time today, I look at it and I tell myself, "I'm... I need to take more steps today	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Data-Driven Decision Making 14
if I haven't taken enough steps that day, I would actually um... deliberately take some, maybe evening walks or deliberately take a longer route err... That doesn't take up too much of time. Maybe, you know, a route that adds that's maybe another five, ten minutes longer. That's about it right and that should usually clock me another two thousand steps or so.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Gamified Goals_Goal completion behavior 5
I will just think back on what could have happened during that period of time. Right, was it because there were some disruptions that woke me up or... or... such right and well, every now and then I do think about on days where I get good sleep score.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Gamified Goals_Goal completion behavior 5
monitoring my heart rate during badminton. Well, not that I'm trying to see if it goes too high and and whether or not I'll just drop on the floor and faint, but rather I am... looking at the intensity of the game and um... also, you know, after... after the sessions, I would look at at which time period was my heart rate the highest	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Performance Monitoring 8

Figure B12: Sample Coding - Self-Quantification, Gamified Goals. Numbers indicate frequency of code application across the dataset.

that tells me which other players [laughter] that I need play more with [laughter] to clock in more brain cells... and even more... puts more, I mean requires more physical output from me	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Social Motivations_Peer observation 3
I think the fact that I've been using it for five years and haven't thought of dropping it... translates to my intention to continue to use this.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Motivational Evolution_Sustained use 3
have been hearing from my friends that, oh, you know, these are that the accuracy are questionable	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Barriers_Accuracy concerns 3
I still believe that all these trackers, the sensors that they use and the apps that they have put in has a certain level of accuracy	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Accountability function 2
I would say... with all the features that are available in the market at this point of time, technology, AI camera, well, if I have an AI camera that I just have to scan and it tells me and it automatically logs um... (the estimated calorie count)	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Trend Analysis 5

Figure B13: Sample Coding - Self-Quantification, Social Motivations, Motivational Evolution, and Barriers. Numbers indicate frequency of code application across the dataset.

if an actual numerical data tells me can give me a better view. I think for me, who's a number person, it would be it would be it would be a fun thing for me to look at.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Data-Driven Decision Making 14
I do know people who you know, looks at the number and "yeah, okay, it's 2000 (steps taken)	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Self-Quantification_Accountability function 2
I don't usually ask for their feedback.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Motivational Evolution_Stage progression 4
but most of the people that I know that has started using fitness band, they have been pretty active to begin with.	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Motivational Evolution_Stage progression 4
they have already been doing it, but and they think that the features offered in this um... wearables and apps can complement or give them better visibility	Experiences and perceptions of adults towards the use of health apps and wearables _ID001_30Sep25_IDITransc (2).pdf	Motivational Evolution_Stage progression 4

Figure B14: Sample Coding - Self-Quantification and Motivational Evolution. Numbers indicate frequency of code application across the dataset.

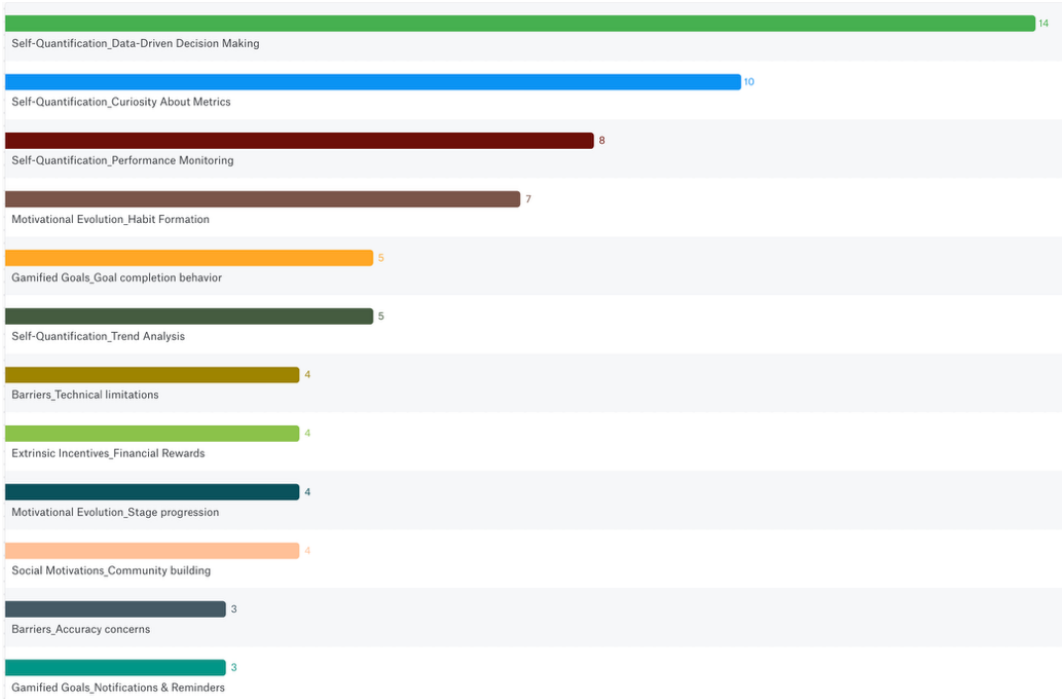


Figure B15: Sample Coding - Comprehensive Dataset View

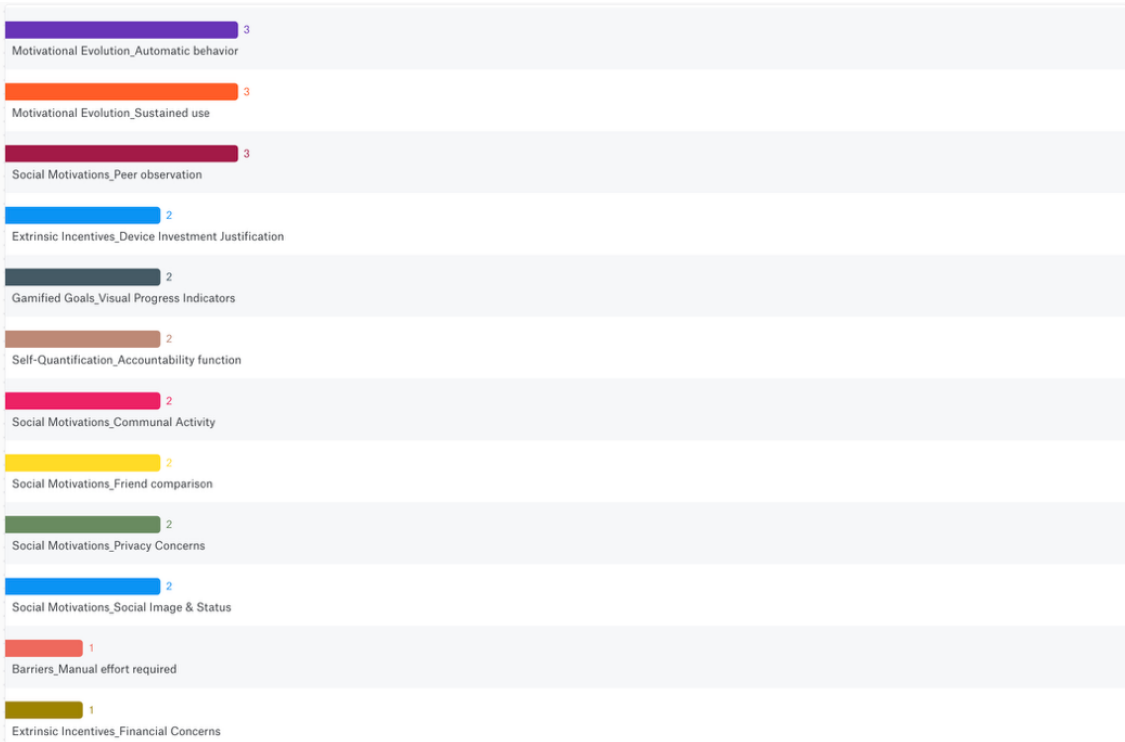


Figure B16: Sample Coding - Comprehensive Dataset View

Figure B17: Sample Coding - Comprehensive Dataset View

APPENDIX C: COREQ CHECKLIST

No.	Item	Guide Questions / Description	Reported?	Location in Manuscript
1	Interviewer/facilitator	Which author(s) conducted the interview or focus group?	Yes	Methods, p.2 "Semi-structured interviews… were conducted"
2	Credentials	Researcher credentials (PhD, MD, etc.)	No	Not reported
3	Occupation	Occupation at the time of the study	Partial	Implied – Graduate students (SPH 5409 module submission)
4	Gender	Researcher's gender	No	Not reported
5	Experience and training	Experience/training of researcher	Partial	Implied – Formal qualitative training (SPH 5409 course)
6	Relationship established	Relationship established prior to study?	No	No prior relationship established
7	Participant knowledge of the interviewer	What did participants know about the researcher?	Partial	Methods, p.2 Informed consent outlines study purpose
8	Interviewer characteristics	Any characteristics reported?	No	Not reported
9	Methodological orientation and theory	Methodological orientation underpinning the study	Yes	Methods, p.2 Hybrid deductive–inductive; cites Fereday & Muir-Cochrane (2006)

No.	Item	Guide Questions / Description	Reported?	Location in Manuscript
10	Sampling	How participants were selected	Yes	Methods, p.2 Purposive sample from Singapore
11	Method of approach	How participants were approached	No	Not reported
12	Sample size	Number of participants	Yes	Methods, p.2; Abstract – "Five adult users"
13	Non-participation	Refusal/dropout numbers & reasons	Yes	Discussion, p.7 One participant declined: possible selection bias
14	Setting of data collection	Where data was collected	Yes	Methods, p.2 Online or in-person (Sept–Oct 2025)
15	Presence of non-participants	Anyone else present?	No	Not reported
16	Description of sample	Important characteristics of sample	Yes	Methods, p.2 Age, devices, duration of tracker use, engagement patterns
17	Interview guide	Questions/prompts; pilot testing	Yes	Methods, p.2 Seven topic areas listed
18	Repeat interviews	Repeat interviews conducted?	No	Not conducted
19	Audio/visual recording	Use of recording	Yes	Methods, p.2 Audio-recorded with consent
20	Field notes	Use of field notes	No	Not reported
21	Duration	Length of interviews	Yes	Methods, p.2 – 27–38 minutes
22	Data saturation	Was saturation discussed?	Yes	Discussion, p.7 – Saturation achieved despite small sample
23	Transcripts returned	Member checking (transcripts returned)?	No	Not conducted / not reported
24	Number of data coders	How many coders?	No	Not specified
25	Description of coding tree	Coding tree or structure provided?	Yes	Methods p.2–3; Appendix A p.9; Appendix B p.10–11
26	Derivation of themes	Pre-identified or data-derived themes?	Yes	Methods p.2; Results p.3 – Mixed deductive/inductive; new theme emerged
27	Software	Software used	Yes	Methods, p.2 Atlas.ti v23
28	Participant checking	Did participants provide feedback on findings?	No	Not conducted / not reported
29	Quotations presented	Participant quotations included & identified?	Yes	Results p.3–6 Many quotations with identifier tags
30	Data & findings consistent	Findings consistent with data?	Yes	Throughout Results & Discussion

No.	Item	Guide Questions / Description	Reported?	Location in Manuscript
31	Clarity of major themes	Major themes clearly presented?	Yes	Results p.3–6 Five themes with subheadings
32	Clarity of minor themes	Minor themes / diverse cases described?	Yes	Results p.5–6 Cross-cutting patterns & unexpected finding